Machine learning

Machine | learning

⇓

Machine Study

mommy

father

mother

son

Ind vs Pak

230/19 235/20

Virat

Shami

Winning

Virat

6x ⇒ 244

machine

data ⇒ Model ⇒ Predict

Bike Price data | Capture ⇒ Model ⇒ Price Predict

Price

Machine learning

Supervised learn

Unsupervised

Reinforcement learn

labeled

tabular

Classification

Regression

Bike price

Brand => Price

Km = Pr

multiple

unlabeled

Cluster

0, 1

Yes / no

Cat, Dog

True, False

Usecase

Env

Agente

Decision

[illegible]

AI

Accuracy

Problem

ML

subfield of AI

DL

subfield of ML

GenAI

subfield DL

data(T)

Answer(T)

data → generate

model

model → usage

Machine learning

Linear Regression

Python

supervised learning

label

y ⇐ Price

Price

depe ⇐

output

ind

xn = x

Brand	Km	age	Pow
TVS	50k	2	150cc

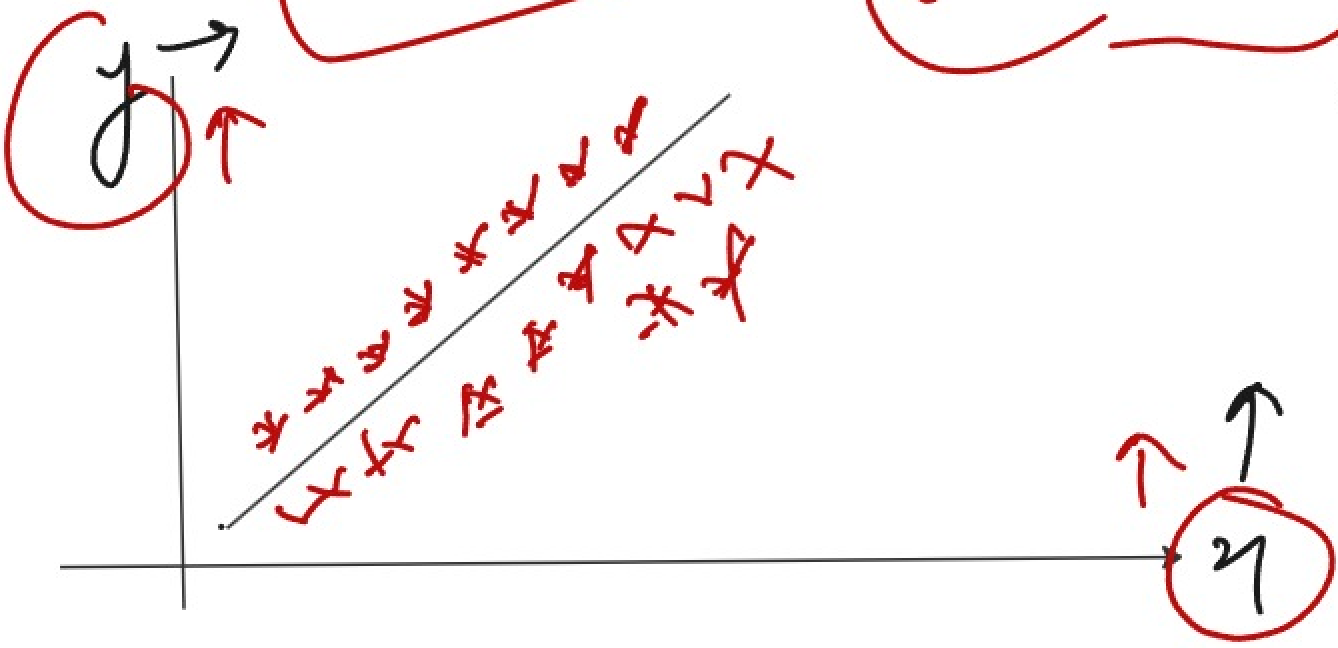
$y = mx + c$

$y \propto x$

y = dependent column

x = independent

- Brand
- km
- age
- Pow



C = 0

M = Slope

Pred => Prediction => $\hat{y}$

Residual ⇐ $(y - \hat{y})$

task1 :- old_Bike_Price

Preprocessing step

dupli..
null
map()
dep..

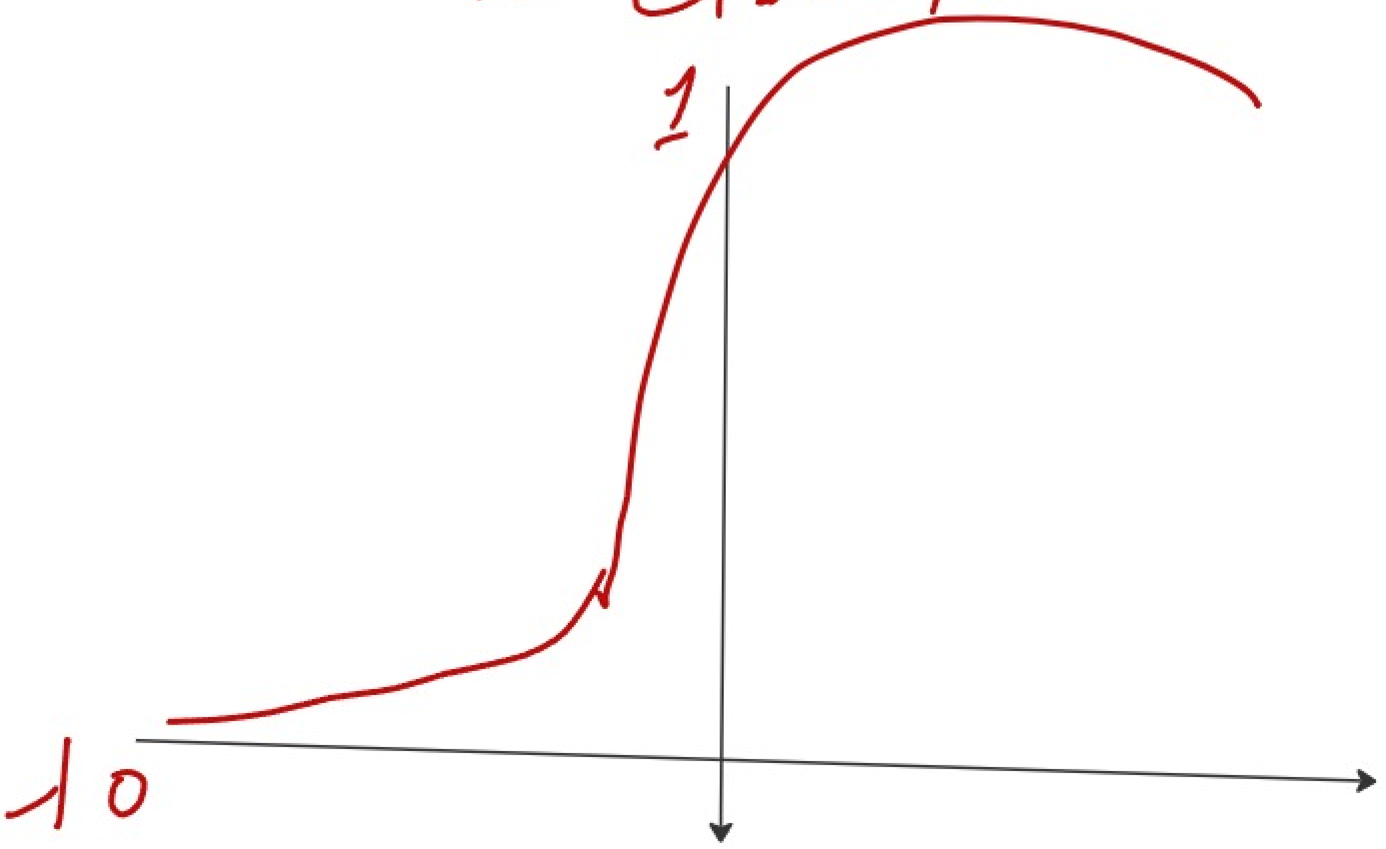
testing

(?)wellcome

logistic (sigmoid)

Sub
label
classification / Regr..

classification



Ensemble learning

M_1 M_2 M_3 M_4 M_5 M_8

Phone

Worth of the Cs

Lr Lasso DT

1 1 0 1

Model

Res

(i) voting

(ii) stacking

(iii) Bagging

(iv) Boosting

Bagging or Boosting

Bagging

DT DT DT DT

(i) Random forest

Boosting

DT DT DT

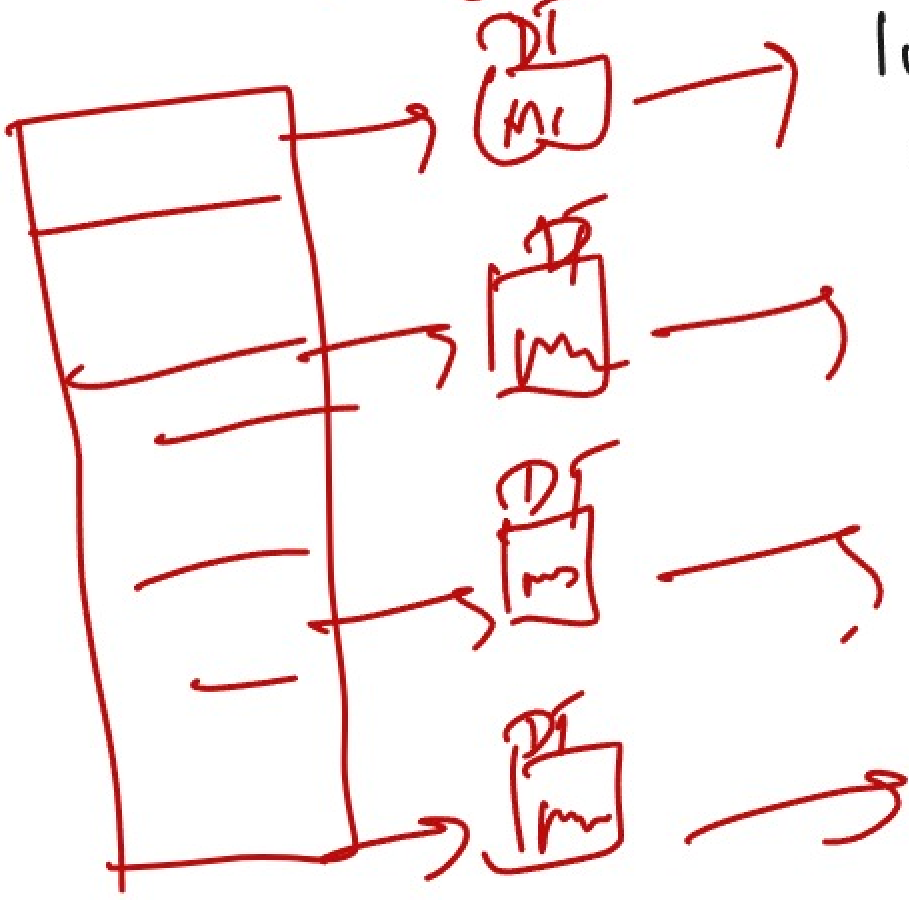
(i) Adaboost

(ii) XGboost

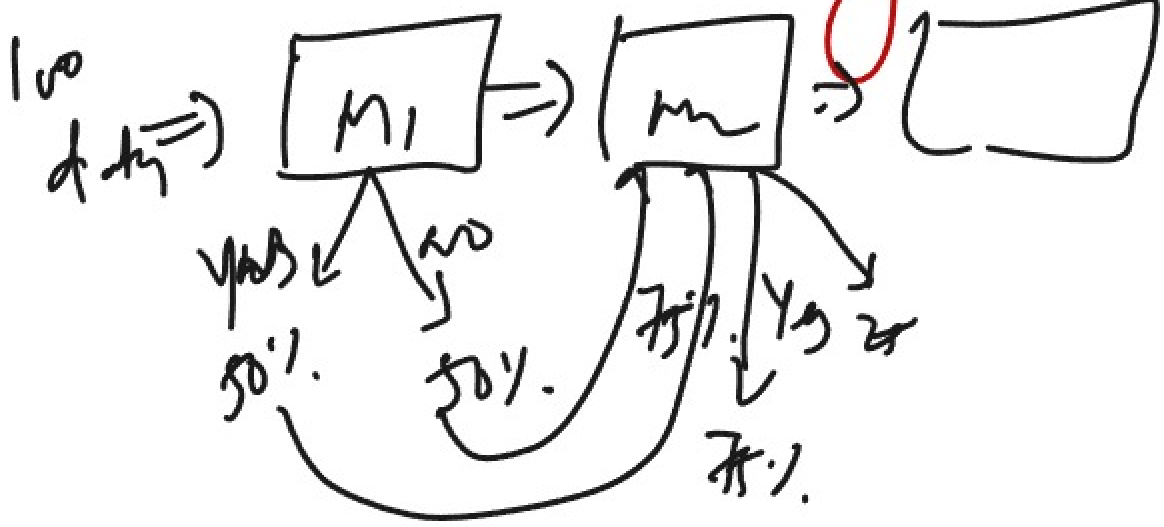
(iii) Gradient descent

Problem

Bagging



Boosting



Each data point learn

Weak learn

Random forest

Supervised Ensemble learning

Classification

Regression

Weak

Random forest => forest => Tree

Naive Bayes

sub, label
classification
Probability — independent $\boxed{H/T}$
— dependent
→ mean

Bayes theorem

A or B

$$P(B|A) = \frac{P(B) \times P(B|A)}{P(A)}$$

B ⇒ independent

A → depen

yes → $\boxed{4.1}$ → 3.5

No → 2

Adding

types of naive Bayes (i) Gaussian
(ii) Bernoulli
(iii) multinomial

flask

framework

Django → web fram

Bike Price Pred.

Price in

flask

model

learn
By
using
data

Price

flask

HTML
CSS

Baki

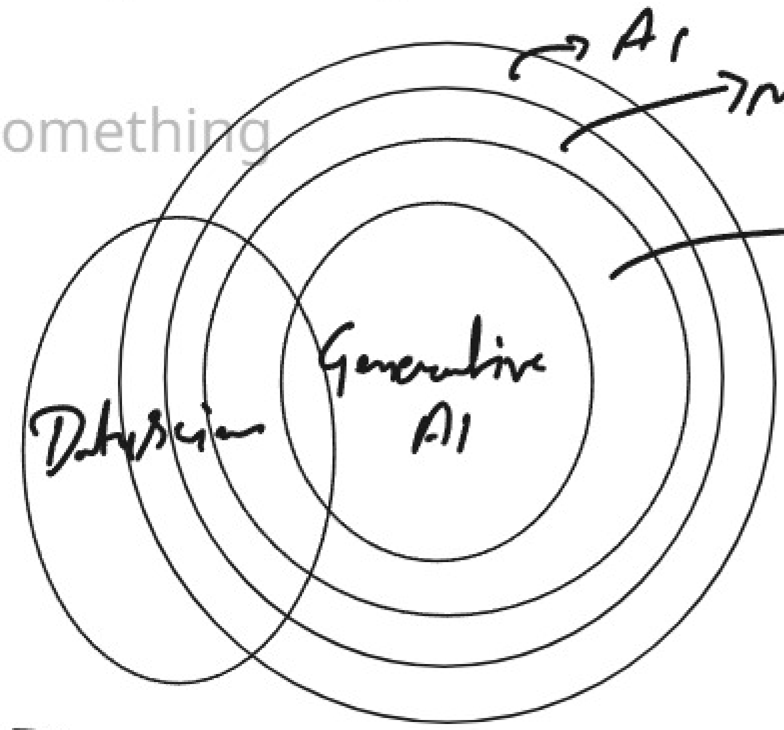
HTML
CSS

Concept of Deep Learning & Neural Network

Choti sister Type something

Big datascience

Junior data science



Audio
Image
vedio
Text
Anim

Deeplearning Algo

(i) ANN
(ii) CNN
(iii) RNN
(iv) LSTM
(v) GRU

forward propagation | feed forward

Input

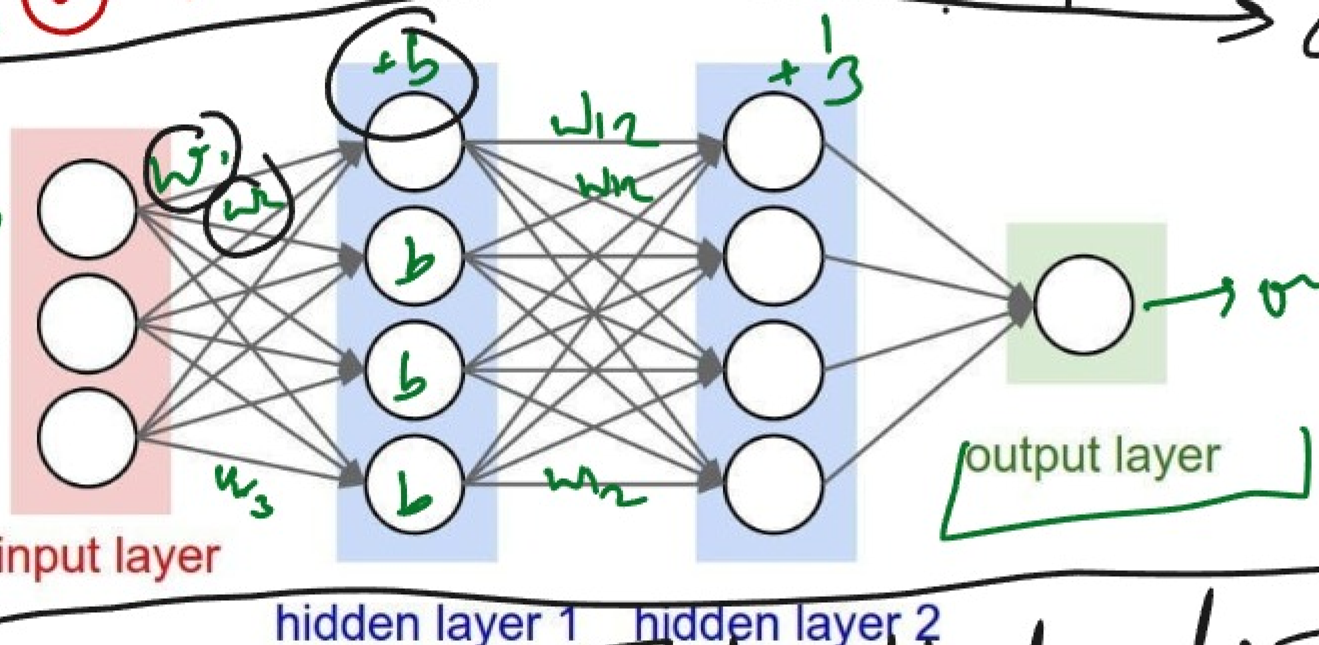
output

Accuracy

x_1

x_2

x_3



output

w, b = -1/b

Backward propagation Bias

Activation function : non linear

types of Acti (7)

(i) Sigmoid / Logistic
(ii) Tanh
(iii) Relu → Ilti
(iv) Eelee Relu
(v) Parametric Relu
(vi) Relu
(vii) Softmax

City
Ray

tensorflow

Python

3.13.34

⇓

3.11.5 version

ML

⇓

Learning rate = 0.1

Deeplearning

ANN
CNN
RNN

ANN tabular

Image, video, Am

Blue? CNN

Cap

CNN

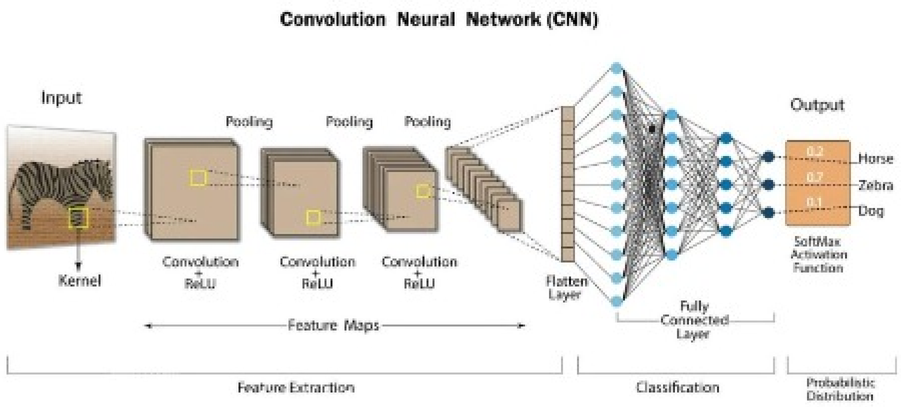
Max →

CNN

(i) Convolution layer

(ii) Pooling layer

(iii) flatten


Convolution Neural Network (CNN)
Input
Kernel
Pooling
Pooling
Pooling
Convolution + ReLU
Convolution + ReLU
Convolution + ReLU
Feature Maps
Flatten Layer
Fully Connected Layer
Output
0.2
Horse
0.7
Zebra
0.1
Dog
SoftMax Activation Function
Feature Extraction
Classification
Probabilistic Distribution

NLP

[Natural language Processing]

Bert

hindi
English
Marwadi

lang

who (r) are
you

Chat

NLP task

(i) Text classification
(ii) Information Retrieval
(iii) Parts of speech tagging
(iv) Language detection & machine translation
(v) Conversion
(vi) Text Summarization
(vii) Topic modelling
(viii) Text Generation

Text Preprocessing

1. LowerCasing
2. Remove HTML tag
3. Remove url
4. Chatword treatment
5. Spelling Correction
4. Remove Stopword
5. handling Emojis
4. Tokenize

hello hi kese https://miro.com/app/board/uXjVI78pAq8=/ ho

who $\boxed{r\ Y}$

Advance
(i) Pos tagging

Deep learning

1. ANN
2. CNN
3. RNN
4. lstm
5. GRU

RNN (Recurrent Neural Network)

ANN → tabular

Accuracy CNN Image, vedio, Animation | machine

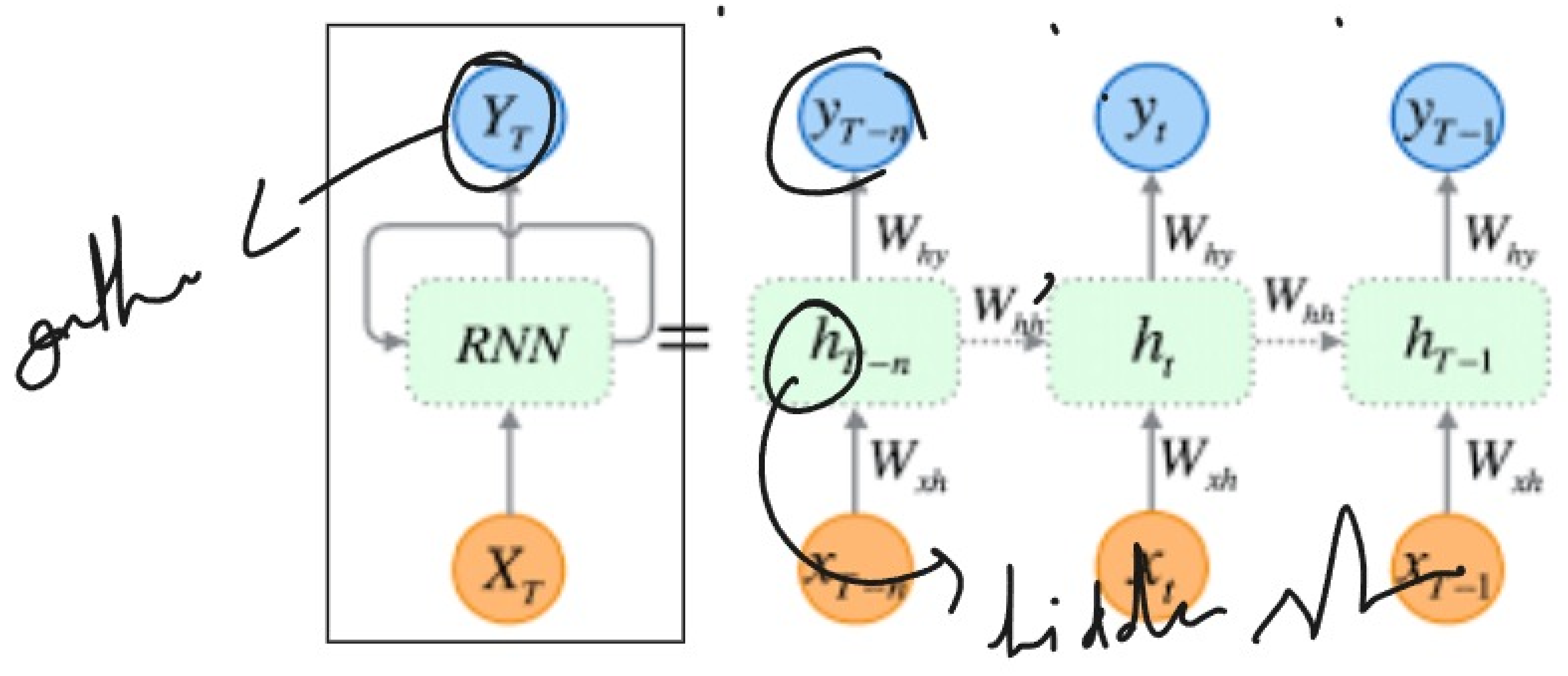
RNN what is your name?

Sequence

where are you from?

Machine learning, ANN, Deep

RNN



What is you

"What" "is" "your" "name" "?"

Types of RNN

(i) one to one RNN (ii) One to many RNN

(iii) many to one RNN (iv) many to many RNN

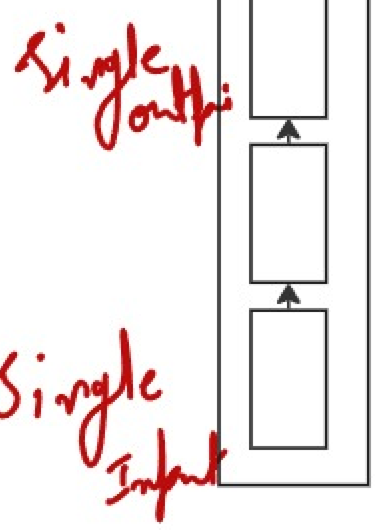
(v) Lstm (vi) GRU

(i) One to one RNN

Binary classification

$\boxed{0 | 1 \,|\, \text{True} \,|\, \text{false}}$

One to one RNN

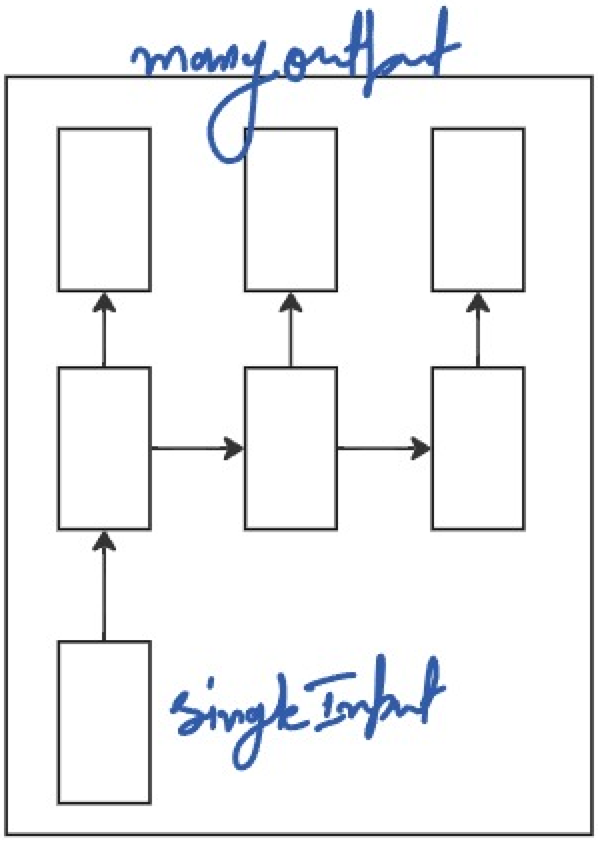


II One to many RNN

Single input => multiple output

? give me a application on fever

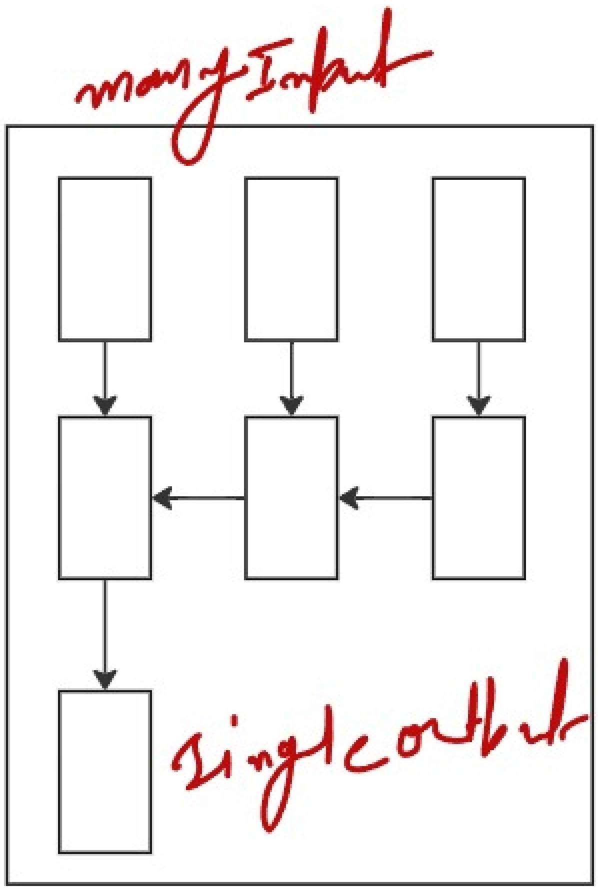
⇓



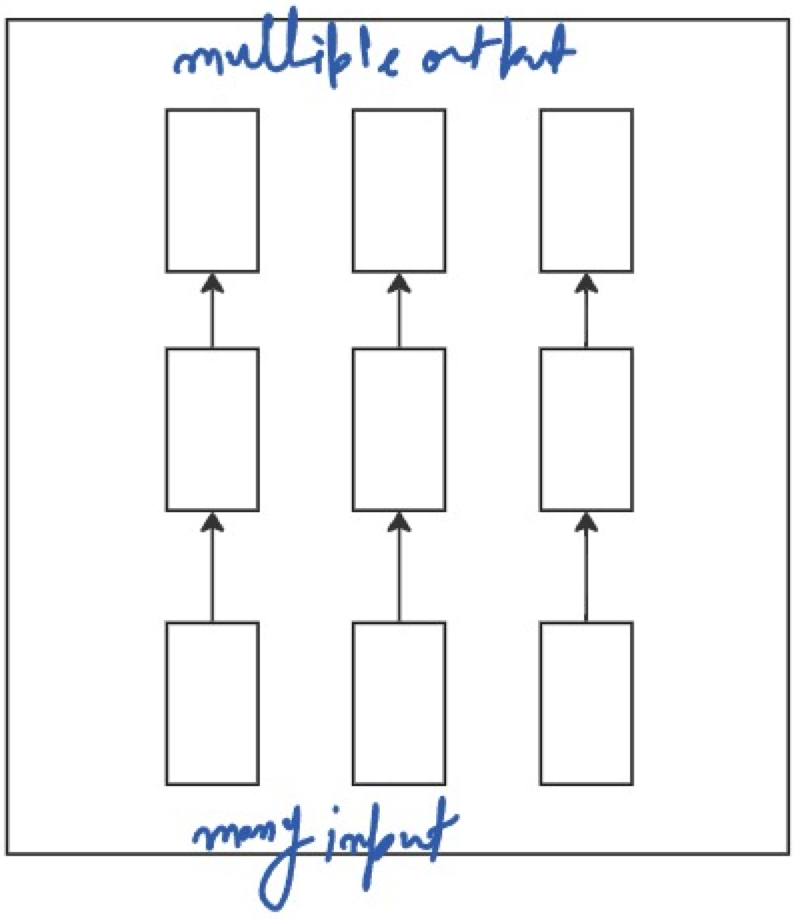
III Many to one RNN

Kya apki tabiyat thik hai

are you okay? Yes sir
No sir



N Many to many RNN



Text vectorization

Word Convert into number

My name is ritik => [1, 0, 2, 3, 4]

(i) BOW (Bag of words)

(ii) TFIDF

(iii) OHE

(iv) N-grams

(v) Custome feature

Bag of words →

order are not matter

Aiyung watch Aiyung

Imagine we have the following two sentences:

1. **"I love dogs."**
2. **"I love cats."**

Create a vocabulary (unique words from all sentences):

- I
- love
- dogs
- cats

Column 1	Column 3	Column 4	Column 5	Column 2
Word	love	dogs	cats	I
Sentence 1	1	1	0	1
Sentence 2	1	0	1	1

So, sentence 1 has I, love, and dogs → vector: **[1, 1, 1, 0]**

Sentence 2 has I, love, and cats → vector: **[1, 1, 0, 1]**

Ad
Simble
Simetric

dis.
out of Vac

This is a good movies
This is not a good movies

N-grams

Types of Ngrams "I Love NLP"

(i) Unigram => ["I", "Love", "NLP"]

(ii) Bigrams => ["I love", "Love NLP"]

(iii) trigram => ["I Love NLP"]

(iv) N grams =>

disa

out of volcubalary
dimention is incresess

TF-IDF (Term Frequency-Inverse Document Frequency)

Suppose we have the following 2 documents:

- **Doc 1:** "I love dogs"
- **Doc 2:** "I love cats"

From both documents, we extract the unique words (just like BoW):

- I, love, dogs, cats

Term Frequency (TF)

Column 1	Column 2	Column 3
Word	TF in Doc 1	TF in Doc 2
I	1/3	1/3
love	1/3	1/3
dogs	1/3	0
cats	0	1/3

: Inverse Document Frequency (IDF)

$$IDF(w) = \log\left(\frac{1}{1 + \text{Number of document Containing word } w}\right)$$

Column 1	Column 2	Column 3
Word	Document Frequency	IDF
I	2	log(2 / (1+2)) = log(2 / 3) ≈ 0
love	2	log(2 / (1+2)) ≈ 0
dogs	1	log(2 / 1) ≈ 0.693
cats	1	log(2 / 1) ≈ 0.693

TF × IDF

Column 1	Column 2	Column 3
Word	TF-IDF in Doc 1	TF-IDF in Doc 2
I	1/3 × 0 = 0	1/3 × 0 = 0
love	1/3 × 0 = 0	1/3 × 0 = 0
dogs	1/3 × 0.693 ≈ 0.231	0 × 0.693 = 0
cats	0 × 0.693 = 0	1/3 × 0.693 ≈ 0.231

✅ Result:

- **Doc 1 vector**: [0, 0, 0.231, 0]
- **Doc 2 vector**: [0, 0, 0, 0.231]

One hot Encoding

Cat dog

Cat = 0

dog = 1

True

False

Lstm and GRU

Advanced version of RNN

Lstm (Long short term memory)

Vanishing gradient

Unit of Lstm

1. forget gate → Remove unnecessary data
2. Input gate → useful add
3. output gate → output
4. cell state → Carries long-term information

forget

Inp.

Advantages of Lstm

1. long sequence
2. Avoids gradient decent problem
3. Audio, lang. speech and time series

Common use Case

Lang. modling, Machine translation

text generate, chats

GRU
(Gated Recurrent Unit)
(2014)

Vanishing gradient
LSTM
Sequence data, Audio, text generation
time series

Component of GRU
(i) Update gate ⇒ (forget + Input)
(ii) Reset gate ⇒ Reset memory than needed
(iii) hidden state As the memory

RNN, LSTM, GRU →

	Column 1	Column 2	Column 3	Column 4
	Feature	RNN	GRU	LSTM
	Gates	None	2 (update, reset)	3 (input, forget, output)
	Memory Handling	Weak	Good	Excellent
	Training Speed	Fast	Faster than LSTM	Slower than GRU
	Complexity	Low	Moderate	High
	Performance	Moderate	High (for many tasks)	Very High (esp. long-term)

GenAi

Deep learning

learning

$\boxed{\text{model}}$

$\Downarrow$

task perform

ANN
CNN
RNN
LSTm
GRU

Machine learning

Machine learning

Machine = model

$\boxed{\text{ML}}$

$\Downarrow$

$\boxed{\text{Data}} \Rightarrow \boxed{\text{Model}} \Rightarrow$ used

$\Downarrow$

EDA, Perprocessing
Clean
Encoding

Generative Ai

Generate Ai

$\Downarrow$ $\boxed{\text{model}}$

Image, text, Audio
vedio

Chatgpt $\Rightarrow$ OPenAI , meta = Meta

Ollama $\Rightarrow$ ollama , Shai =

GeminiApi $\Rightarrow$ Geminiapi .

Encoder and Decoder Based Architecture understanding

Encoder & Decoder

transformer

Encoder

What is your name → Model → [0,1,0,1,

Map()
labelencoding
onehot encoding

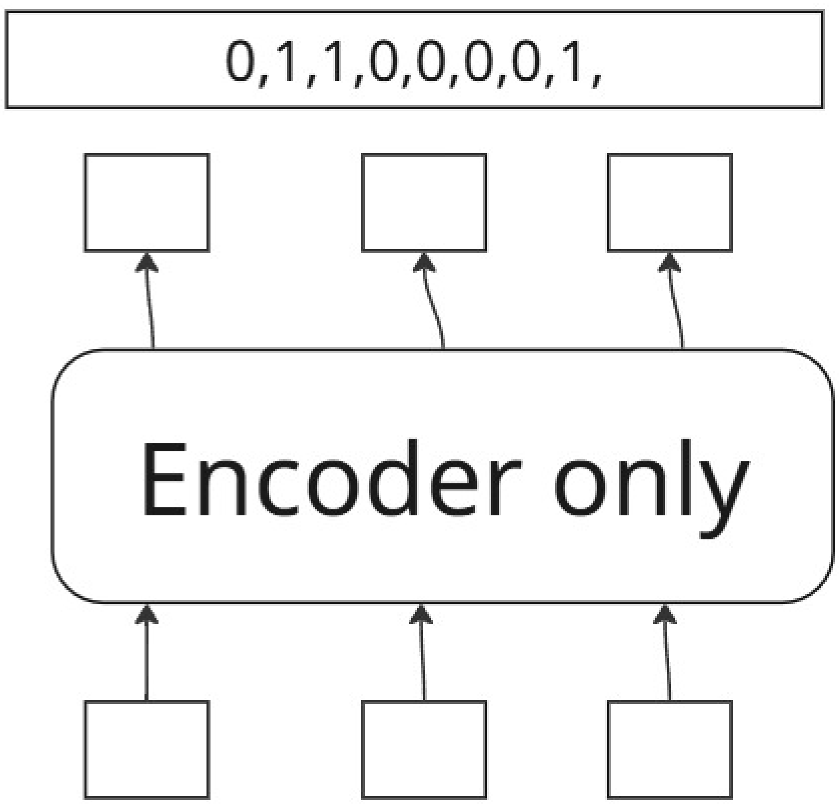
Decoder

decoder

Types of transformer => 3

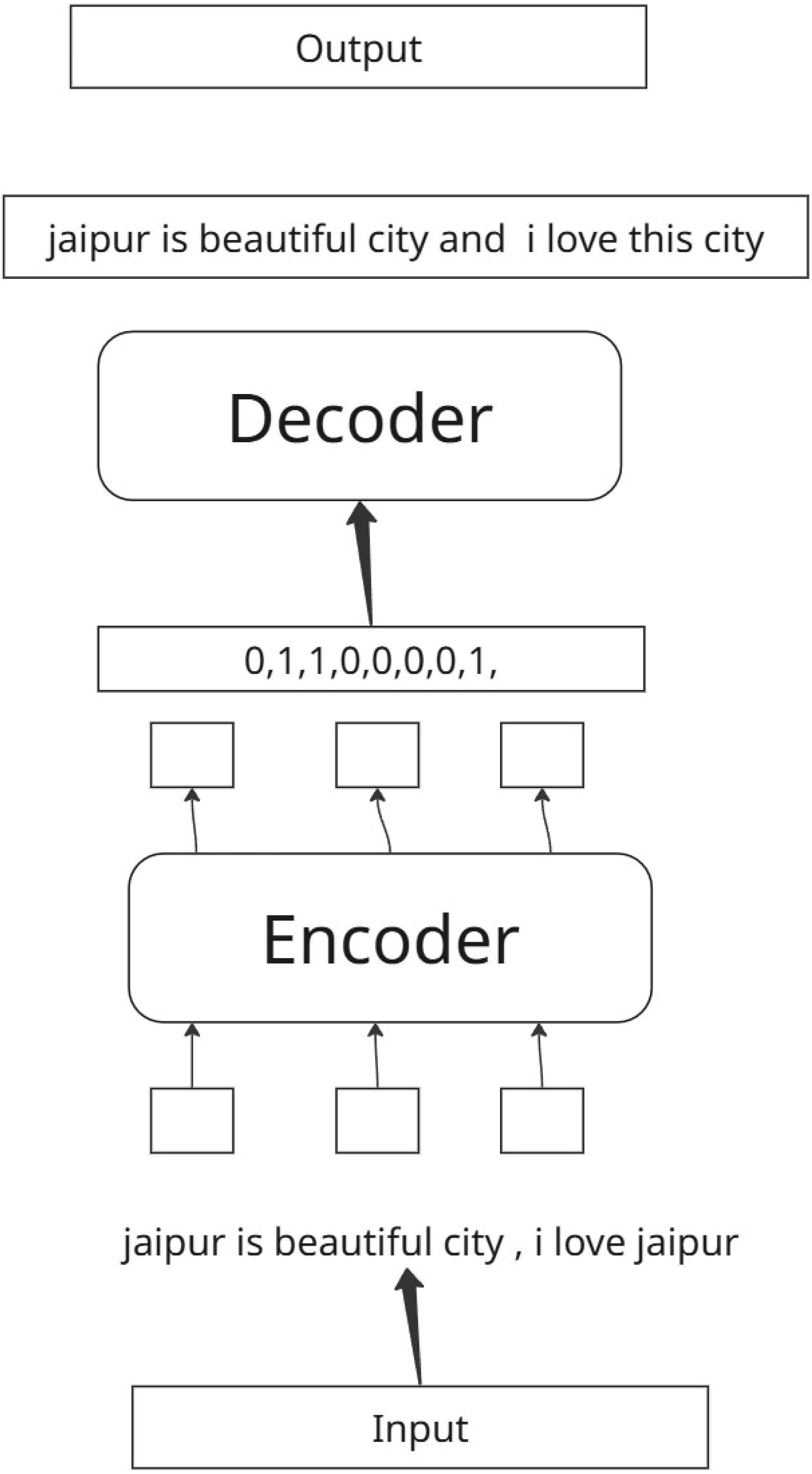
Encoder only
Encoder-decoder
decoder only

Encoder only

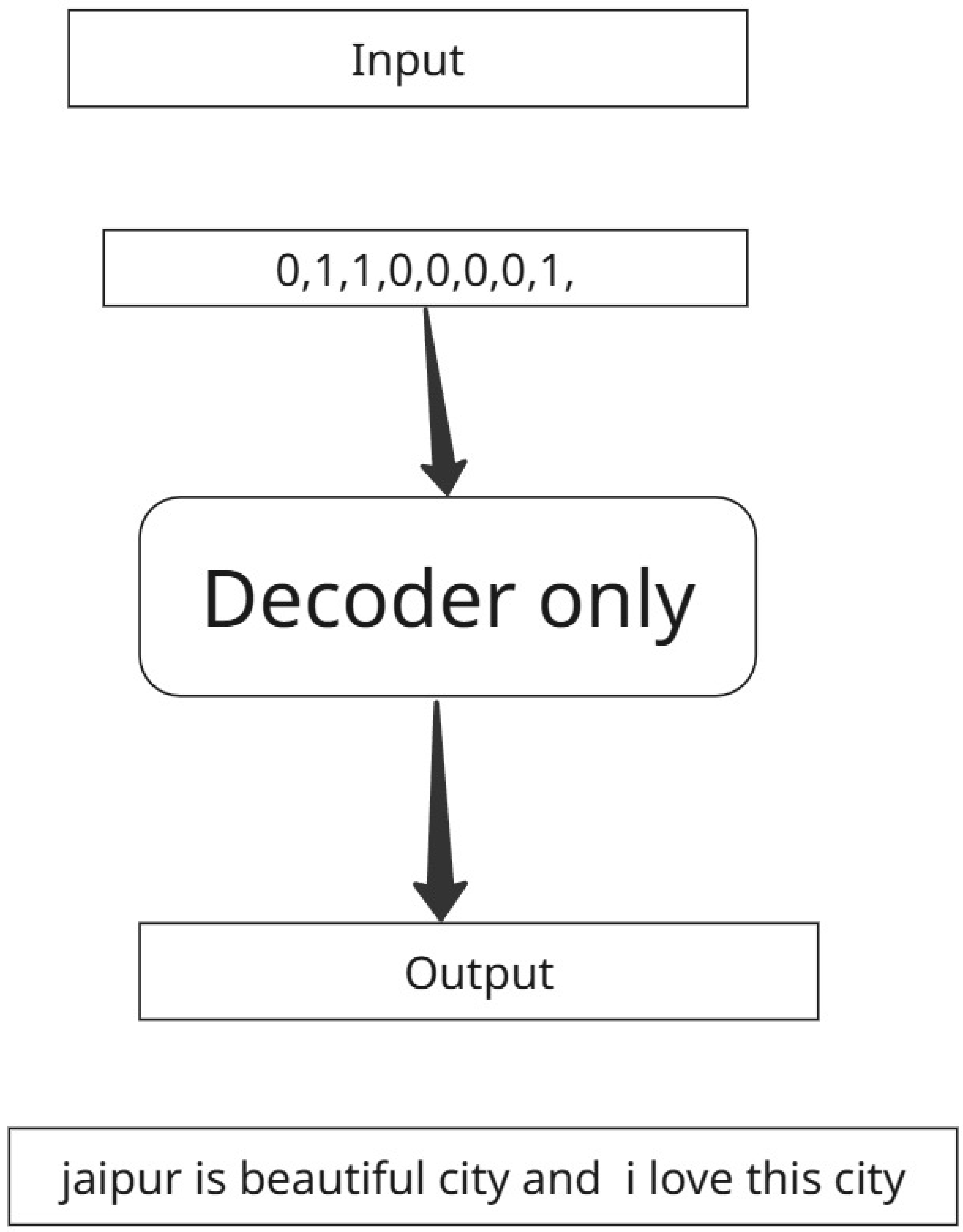


jaipur is beautiful city , i love jaipur

Encoder and Decoder



Decoder only



Bert and Gpt

Bidirectional Encoder Representations From Transformers

Developed by Google (2018)

Raj when to the Amazon forest
kapil join Amazon as developer

Model :- 1.bert best 2. Bert large 3. Bert tinny 4. Bert mini
5.Bert small 6. Bert medium

bert fine tuning

mlm (masked lang model)

Bert input Embedding

1. Token Embedding
2. Segment Embedding
3. Positional Embedding

Gpt (Genrative Pritrain Transformer)

gpt 2

gpt 3

gpt 4